

Sports Injury Risk Detection System

Mathematical Formulas, Weighting Rules & Thresholds

1. Biomechanical Risk

Seven component risks are combined on a 0–100 risk scale.

Biomechanical Risk = (Knee Risk × 0.30) + (Hip Risk × 0.20) + (Trunk Risk × 0.15) + (Landing Risk × 0.15) + (Balance Risk × 0.10) + (Alignment Risk × 0.05) + (Posture Risk × 0.05)

Weights sum to 100%.

2. Redistributed Overall Risk Weights

Fatigue is disabled, so its weight is 0% and the remaining domains use the redistributed weights.

Risk domain	Weight
Biomechanical deviations	38.89%
Previous injury history	22.22%
Movement asymmetry	22.22%
Training load	16.67%
Fatigue	0%

Overall Injury Risk = (0.3889 × Biomechanical Risk) + (0.2222 × Historical Injury Risk) + (0.2222 × Movement Asymmetry Risk) + (0.1667 × Training Load Risk)

Final score is constrained to [0, 100].

3. Risk Classification

Score	Risk level
0–24.99	Low Risk
25–49.99	Moderate Risk
50–74.99	High Risk
75–100	Critical Risk

4. Movement Quality

Movement Quality = 100 – Biomechanical Risk

Higher movement quality corresponds to lower biomechanical deviation risk.

5. Biomechanical Efficiency

Biomechanical Efficiency = (Movement Quality × 0.45) + (Movement Consistency × 0.35) + (Balance × 0.20)

6. Overall Athlete Health Score

A broader health-oriented indicator using biomechanics, asymmetry, injury history, and training load. Higher = better.

Health Score = Weighted(Biomechanics Health, Asymmetry Health, Injury-History Health, Training-Load Health)

Components should be normalized to a common 0–100 health scale before weighting.

Sports Injury Risk Detection System — Component Formulas

Biomechanical, COM, knee-valgus, trunk, and hip-stability calculations

7. Approximate Center of Mass (COM)

Segment weights sum to 1.00:

Segment	Weight
Head	0.08
Trunk	0.48
Thighs	0.24
Lower legs	0.12
Feet	0.08

Approximate COM = (0.08 × Head) + (0.48 × Trunk) + (0.24 × Thighs) + (0.12 × Lower Legs) + (0.08 × Feet)

The trunk is approximated by the midpoint between shoulder center and hip center.

Trunk Center = (Shoulder Center + Hip Center) / 2

8. Knee Valgus Risk

Knee Risk = (0.70 × Worse-Side Risk) + (0.30 × Better-Side Risk)

MediaPipe 2D normalized medial displacement is relative to leg length.

Displacement ratio	Category
< 0.05	Normal
0.05–0.10	Mild
0.10–0.15	Moderate
≥ 0.15	High

9. Trunk Lean Risk

Trunk Score = 100 – [(Average Trunk Lean × 3) + (Extra Penalty if Maximum Lean > 15° × 2)]

Trunk Risk = 100 – Trunk Score

Category	Threshold
Normal	< 5°
Mild	5°–<10°
Moderate	10°–<15°
Severe	15°–<20°
At/above severe threshold	≥ 20°

10. Hip Stability Risk

Instability = (0.35 × Tilt Component) + (0.25 × Variability Component) + (0.20 × Displacement Component) + (0.20 × Symmetry Penalty)

Hip Score = 100 – Instability

Hip Risk = 100 – Hip Score = Instability

Thus, when both use the same 0–100 scale, hip risk can be represented directly by instability.

11. Movement Asymmetry Thresholds

Difference	Category
< 10%	Optimal
10%–<15%	Mild
15%–25%	Moderate
> 25%	Severe

Sports Injury Risk Detection System — Risk Models

Alignment, training load, and historical injury scoring

12. Joint Alignment Risk

Component weights:

Joint component	Weight
Knee	0.30
Hip	0.25
Ankle	0.15
Shoulder	0.15
Elbow	0.15

Alignment Score = (0.30 × Knee Score) + (0.25 × Hip Score) + (0.15 × Ankle Score) + (0.15 × Shoulder Score) + (0.15 × Elbow Score)

Alignment Risk = 100 – Alignment Score

Component weights sum to 1.00.

13. Training Load Risk Thresholds

Training-load condition	Risk interpretation
40–70	Optimal → Low Risk
>70–85	Moderate fatigue / overreach → Moderate Risk
>85	High fatigue / spike → High Risk
<25 + high intensity	Under-conditioning → Moderate Risk

14. Historical Injury Risk Scoring

Per-injury base score:

Base Injury Score■ = (0.35 × Status■) + (0.35 × Severity■) + (0.20 × Recency■) + (0.10 × Biomechanical Relevance■)

The four weights sum to 100%.

15. Recency Decay

Recency■ = 100 × exp(–Years Since Injury■ / HISTORY_RECENCY_HALF_LIFE_YEARS)

Equivalently: Recency■ = 100 × e^(–Years Since Injury■ / HISTORY_RECENCY_HALF_LIFE_YEARS).

16. Multiple-Injury Aggregation

Injuries are sorted by severity. Secondary injuries are damped using HISTORY_AGGREGATION_DECAY raised to their index.

Adjusted Score■ = Base Injury Score■ × (HISTORY_AGGREGATION_DECAY)■

S_{hist} = 100 × [1 – ∏_{i=0}ⁿ⁻¹ (1 – Adjusted Score_i/100)]

This bounded complement aggregation keeps historical risk within 0–100.

17. End-to-End Scoring Flow

Component risks → Biomechanical Risk → Movement Quality → Redistributed Domain Weights → Overall Injury Risk → Risk Category

The scoring system is transparent and rule-based: each score can be traced to a formula, weight, threshold, or historical-injury factor.